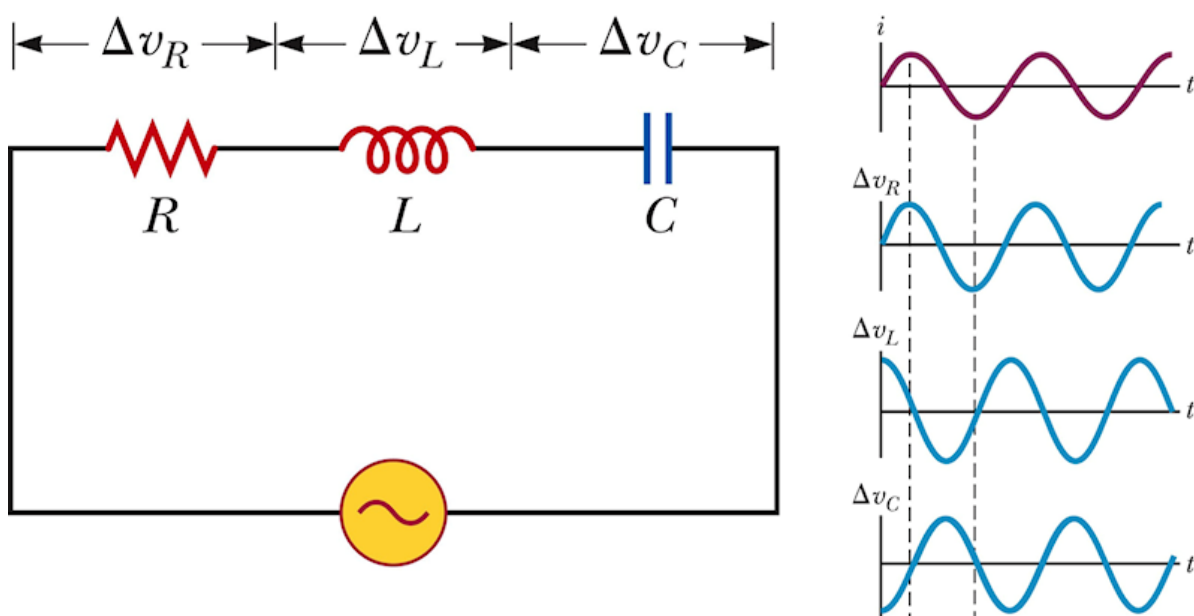




## **Instruction Manual for Electrical Technology Laboratory** **Department of Electrical Engineering, IIT (ISM) Dhanbad**

Expt. No: 04

### **STUDY OF A SERIES R-L-C CIRCUIT**



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Experiment No.

**TITLE: Study of a series R-L-C circuit**

**OBJECTIVE:** To draw the phasor diagram using experimental data and therefrom, to find out circuit constants of a resistor, a coil and a condenser connected in series.

**APPARATUS USED:**

| Serial No. | Name of the Equipment | Rating | Model No. of the Equipment | Makers Name |
|------------|-----------------------|--------|----------------------------|-------------|
| 1.         |                       |        |                            |             |
| 2.         |                       |        |                            |             |
| 3.         |                       |        |                            |             |
| 4.         |                       |        |                            |             |

**CIRCUIT DIAGRAM:**

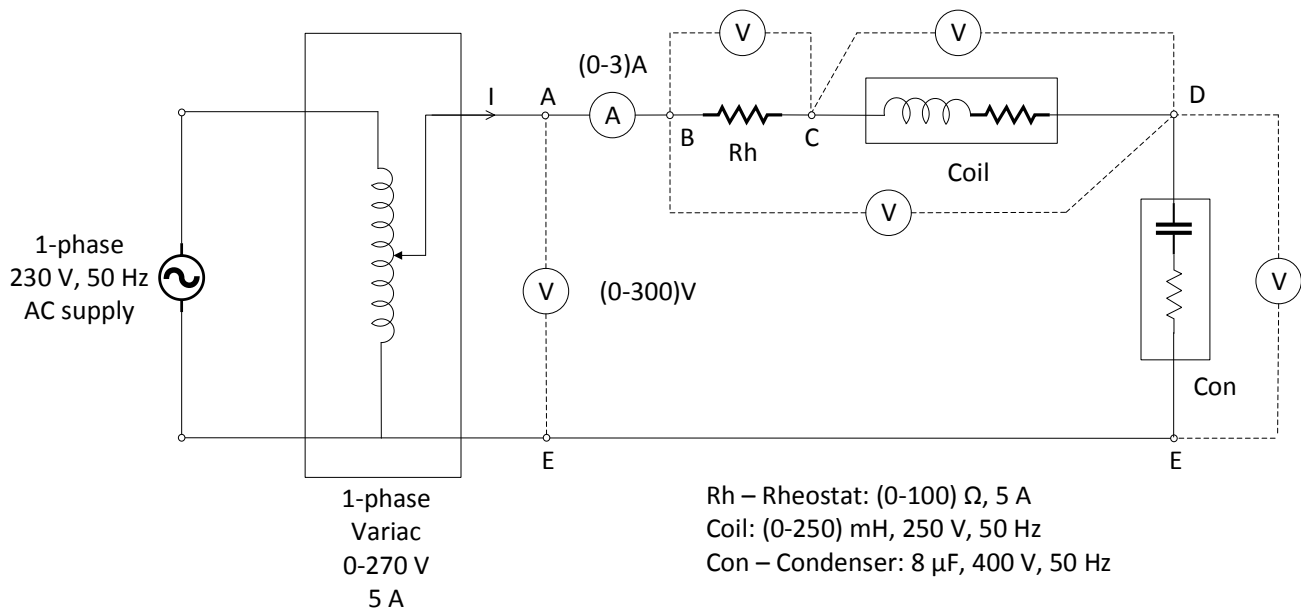


Fig.1. Circuit diagram for a series R-L-C circuit. The dotted connection lines indicate the temporary connections of a multimeter in AC voltmeter mode.

## PROCEDURE:

- 1) Connect the circuit as per the circuit diagram shown in Fig. 1 and be sure that the variac is set at 0 volt.
- 2) Switch on the power supply to the variac and adjust the variac to set a suitable voltage.
- 3) Measure the current passing through the circuit by using ammeter and the voltages across various components as mentioned in the Table I, by using a multimeter in AC voltage mode.
- 4) Take three sets of readings for different values of the voltage applied to the circuit.
- 5) Draw the phasor diagram to scale in a graph paper, for each set of readings and, therefrom, find out the electrical circuit constants of each component, as shown in the Table II, by averaging the values from the three sets of readings.

### **Procedure to draw the Phasor Diagram** (Observe Fig. 2. carefully)

Draw  $\overrightarrow{OM}$  to represent the current phasor  $\bar{I}$ . Draw  $\overrightarrow{OP}$  in the same line as  $\overrightarrow{OM}$  with length equal to  $V_{BC}$ . Draw an arc in the upper side of  $\overrightarrow{OM}$ , with centre at P and radius equal to  $V_{CD}$ . Draw another arc centred at O with radius of  $V_{BD}$  to intersect the previous arc at Q. Draw  $\overrightarrow{QR}$  perpendicular to  $\overrightarrow{OM}$ . Then  $\overrightarrow{RQ}$  and  $\overrightarrow{PR}$  represent the voltage across the inductance and internal resistance of the coil, respectively. Draw an arc below  $\overrightarrow{OM}$  with centred at Q with radius equal to  $V_{DE}$  and another arc centred at O with radius  $V_{AE}$  to intersect each other at S. Draw  $\overrightarrow{QV}$  parallel to  $\overrightarrow{OM}$  and  $\overrightarrow{ST}$  perpendicular to  $\overrightarrow{QV}$ . Then  $\overrightarrow{TS}$  and  $\overrightarrow{QT}$  represent the voltages across the capacitance and the internal resistance of the condenser. Then calculate the circuit constants mentioned in the Table III from the phasor diagram.

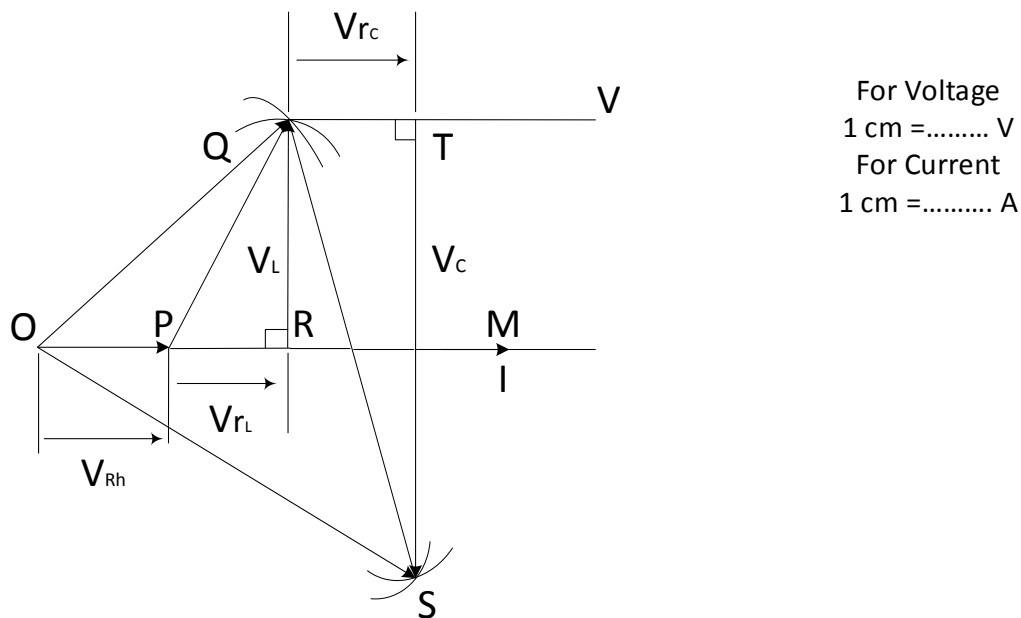


Fig. 2. Phasor diagram for a series R-L-C circuit

Table III

| SL. No. | Circuit constant             | Symbol   | Formula                                 | Unit     |
|---------|------------------------------|----------|-----------------------------------------|----------|
| 1       | Resistance of the Resistor   | $R_{th}$ | (OP volt) / (OM amp)                    | $\Omega$ |
| 2       | Resistance of the Coil       | $r_L$    | (PR volt) / (OM amp)                    | $\Omega$ |
| 3       | Inductance of the Coil       | L        | (RQ volt) / ( $100\pi$ rad/sec. OM amp) | H        |
| 4       | Resistance of the Condenser  | $r_c$    | (QT volt) / (OM amp)                    | $\Omega$ |
| 5       | Capacitance of the Condenser | C        | (OM amp) / ( $100\pi$ rad/sec. TS volt) | F        |

### COMMENTS/ DISCUSSIONS:

Write your comments on the results obtained and discuss the discrepancies, if any.

### PRECAUTIONS:

- 1) Turn off the power switch to equipment before making connections.
- 2) Don't use broken connecting wires.
- 3) Maintain a work space clear of extraneous material such as books, papers, and clothes.
- 4) Don't wear loose clothes.
- 5) Wear shoes that cover the feet.

### REPORT:

- 1) Observe whether a voltage across any components is more than the voltage applied to the circuit.
- 2) What is resonance in an electrical circuit? Find the frequency for which the above circuit will be resonant.
- 3) Discuss the phenomenon of series resonance in an electrical circuit.
- 4) Is it possible to have a voltage drop across the energy storage element greater than the supply?

## DATA SHEET

Experiment No. \_\_\_\_\_

Date: \_\_\_\_\_

TITLE: Study of a series R-L-C circuit

Name: \_\_\_\_\_ Sem: \_\_\_\_\_ Year: \_\_\_\_\_

Adm. No. \_\_\_\_\_ Branch: \_\_\_\_\_

Experimental Data: -

Table I

| SL. No. | I (A) | V <sub>BC</sub> (V) | V <sub>CD</sub> (V) | V <sub>BD</sub> (V) | V <sub>DE</sub> (V) | V <sub>AE</sub> (V) | Remark (if any) |
|---------|-------|---------------------|---------------------|---------------------|---------------------|---------------------|-----------------|
|         |       |                     |                     |                     |                     |                     |                 |
|         |       |                     |                     |                     |                     |                     |                 |
|         |       |                     |                     |                     |                     |                     |                 |

Table II

| SL. No.        | V <sub>AE</sub> (V) | Resistance of Resistor ( $\Omega$ ) | Inductance of the Coil (H) | Internal Resistance of the Coil ( $\Omega$ ) | Capacitance of the Condenser (F) | Internal resistance of the Condenser ( $\Omega$ ) |
|----------------|---------------------|-------------------------------------|----------------------------|----------------------------------------------|----------------------------------|---------------------------------------------------|
|                |                     |                                     |                            |                                              |                                  |                                                   |
|                |                     |                                     |                            |                                              |                                  |                                                   |
|                |                     |                                     |                            |                                              |                                  |                                                   |
| Average Values |                     |                                     |                            |                                              |                                  |                                                   |

\_\_\_\_\_  
(Signature of the teacher)